MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. A normal to $2x - 3y + 1 = 0$ is
   - A. $(3, 2)$
   - B. $(2, -3)$
   - C. $(2, 3)$
   - D. $(1, 0)$
2. A direction vector of $x + 2y = 0$ is
   - A. $(-2, 1)$
   - B. $(1, 2)$
   - C. $(2, 1)$
   - D. $(0, 2)$
3. Centre of $(x + 2)^2 + (y - 3)^2 = 4$:
   - A. $(2, 3)$
   - B. $(-2, -3)$
   - C. $(3, -2)$
   - D. $(-2, 3)$
4. Radius of $x^2 + y^2 = 9$:
   - A. 9
   - B. 81
   - C. 3
   - D. $\sqrt{3}$
5. $x^2 + y^2 = -1$ describes
   - A. a circle
   - B. the empty set
   - C. the origin
   - D. a line

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. Projection of $(3, 4)$ onto the x-axis:
   - A. $(0, 4)$
   - B. $(0, 0)$
   - C. $(3, 0)$
   - D. $(4, 3)$
2. Distance from $(1, 5)$ to $y = 2$:
   - A. 3
   - B. 5

C. 1
D. $\sqrt{10}$

3. For $ax + by + c = 0$ to define a line, we need
   A. $a \neq 0$ alone
   B. $b \neq 0$ alone
   C. $c \neq 0$
   D. $(a, b) \neq (0, 0)$
4. $x^2 + y^2 = 0$ describes
   A. a circle of radius 1
   B. the single point O
   C. a line
   D. the plane
5. Circle $x^2 + y^2 = 4$ and line $x = 2$: number of common points
   A. 0
   B. 2
   C. 1
   D. infinitely many

True or false? Justify.

1. The orthogonal projection of P onto a line is unique.
2. Every equation $x^2 + y^2 + c = 0$ describes a circle of positive radius.
3. A line can meet a circle at three distinct points.

> **Key point** · **Self-assessment.** I give a nonzero normal; I check the projection; I complete both squares; I distinguish circle, point and empty set.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.